

It is very unfortunate that the burden of such hazard processing is passed on from the most developed world to the most marginalised communities of developing countries like India. For example, CRTs, a Basel listed waste, is being openly traded between the United States and India as CRT monitors have become obsolete with the advent of flat-screen monitors. It costs \$20 to recycle a single computer in the US, while the same could be recycled in India for only \$2—a saving of \$18 if the same is exported to India. This trade, though flourishing, is illegal and contravenes the international Basel treaty that prohibits trans-boundary movement of hazardous goods and products.

The Basel Convention

The Basel Convention of 1992, to which India is a signatory, is an international treaty whose central goal is environmentally sound management. It involves controlling hazardous waste from its production to storage, transport, reuse, recycling and final disposal. It also includes prevention and monitoring of illegal traffic of hazardous waste and promotion of cleaner technologies.

The present Indian Hazardous Waste Rule defines wastes as substances that need to be controlled in their movement, disposal, recovery or reclamation and recycling operation. However, this rule (Basel Convention & Hazardous Waste Rule) does not completely ban the movement of hazardous wastes and is more in nature of ensuring controlled movement. Non-implementation, poor regulations and the absence of a clear policy of central and state governments in dealing with e-waste add to the problems.

Effects of e-waste constituents on health

Environmental pollution is an act of introducing extraneous substances or energy into the environment that induce unfavourable changes. These changes may affect man directly or indirectly by endangering health, harm-

Toxic Elements present in a Computer System*

Elements	Quantity
Plastic	7,240 kg
Lead	1,980 kg
Barium	9,920 kg
Beryllium	4,940 gm
Cadmium	2,961 gm
Chromium	1,980 gm
Mercury	0.693 gm
Arsenic	0.409 gm

*Weighing approximately 31.5 kg

ing living resources and ecosystem, or interfering with legitimate uses of the environment.

Potentially toxic and relatively accessible chemicals and metals are present at less than 0.1 per cent level in the earth's crust, but anthropogenic activities such as individual processing of metals and metallic compounds add up to the natural background levels of these elements. Most of the electrical and electronic products contain significant quantities of toxic metals and chemicals. If these are left untreated to lie around in landfills or dumps, these contaminate the soil and after leeching through the soil, pollute subsoil aquifers, posing obvious health hazards to humans.

The ill-effects of various toxic waste materials are discussed below.

Antimony. Antimony (Sb) is a metal with a variety of industrial applications including as a flame retardant and as a trace component of metal solders. In some forms, antimony shows many chemical similarities to arsenic, including its toxicity. Exposure to high levels of antimony can lead to severe skin and related health problems.

Barium. Barium (Ba) is a very soft silvery-white metal that is used in computers in the front panel of the cathode ray tube (CRT) to protect the users from radiations. Long-term exposure to barium may cause brain swelling, muscle weakness and damage to liver and spleen.

Beryllium. Beryllium (Be) is the

lightest alkaline earth element that has a low atomic number, high ionisation energy and great electro-negativity characteristics. It is commonly used and found on the motherboards and finger chips as a copper-beryllium alloy to strengthen the tensile strength of connectors and plugs while maintaining electrical characteristics. Because of the absence of homeostatic control mechanism, it tends to accumulate in the human body without being excreted. Basically, beryllium is mutagenic and carcinogenic. Its retention within the lungs leads to the development of a disease condition called 'Berylliosis.'

Brominated flame retardants. Brominated flame retardants (BFRs) are used in plastic housings of electronic equipment and in circuit boards to prevent flammability. More than 50 per cent of BFR usage in the electronics industry consists of tetra bromo bisphenol-A, polybrominated biphenyl ethers and polybrominated biphenyls, which are believed to cause cancer of the digestive and lymph systems.

Cadmium. Cadmium (Cd) is a toxic heavy metal belonging to the same family as zinc and mercury. It is relatively rare but toxic to all systems and functions at high levels of exposure. It is used in formulation of alloys, electroplating, SMD chip resistors and Ni-Cd batteries. Cadmium stearates are used as stabilisers in the production of polyvinyl chloride (PVC) plastics.

Cadmium is a potent enzyme inhibitor and is known to interact with the sulph-hydryl (-SH) groups of several enzymes. Symptoms of cadmium intoxication include hypertension, respiratory disorders, and damage to liver and kidney.

Copper. Copper (Cu) is an essential trace element that is ubiquitous in the earth's crust. It is tough, soft, highly resistant to corrosion and second only to silver (and gold) in thermal and electrical conductivities. Therefore it finds extensive use in thermal and electrical conductor tracks of printed circuit boards. Large depositions of copper salts in the body may cause hemolytic,